9b. Soft Interventions and σ-Calculus

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Soft-Intervention and σ-Calculus* (Prof. Sanghack Lee, GSDS SNU; adapted from Elias Bareinboim)

The central question is how to evaluate realistic policies, which change a variable's mechanism to a conditional or stochastic rule rather than fixing it to a constant, and how the do-calculus generalises to such soft interventions.

These notes consolidate the slides, define the canonical types of regimes, work the reducible example, explain where do-calculus intuition breaks, state the σ-calculus rules alongside their do-calculus counterparts, and identify soft-intervention effects with c-factors. Physical PDF page numbers are listed in Section 7. Reference: Correa and Bareinboim (2020), "A calculus for stochastic interventions", AAAI.

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1 Motivation

Purpose. Show why atomic $\text{do}(x)$ interventions are often not the policies we can implement.

A tutoring programme. For each student we observe the previous GPA W, motivation Z (low/high), tutoring X, and the final GPA Y. Motivation depends on previous GPA (and unobserved factors); students get tutoring depending on motivation; the final GPA depends on previous GPA, motivation and tutoring: $W \to Z \to X \to Y$, $W \to Y$, $Z \to Y$.

With large data and ML, $P(y \mid w, z, x)$ can be predicted with small error. But this distribution reflects the current, *natural regime* (regime: the mode a system operates in). Decisions intervene on the regime, so we must predict GPA under a hypothetical, unrealised regime.

What we know so far. From $P(W, Z, X, Y)$ and $\mathcal{G}$ we can identify

$$\begin{aligned} P(y \mid \text{do}(x)) &= \sum_z P(y \mid \text{do}(x), z) P(z \mid \text{do}(x)) && \text{condition on } Z \\ &= \sum_z P(y \mid \text{do}(x), z) P(z) && \text{Rule 3: } (Z \perp_d X)_{\mathcal{G}_{\overline{X}}} \\ &= \sum_z P(y \mid x, z) P(z) && \text{Rule 2: } (Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}}. \end{aligned}$$

Suppose $P(Y = 1 \mid \text{do}(X = 1)) > P(Y = 1)$: we predict improvement if $\text{do}(X = 1)$ is implemented. But what does $\text{do}(X = 1)$ mean in the real world? Mandatory tutoring for every single student. Is this realisable?

Implementing do-interventions. Even when the effect is identifiable, resources may be insufficient (not enough teachers and hours), and effectiveness cannot be guaranteed (non-compliance). For practicality we care about realisable interventions:

Do-like intervention	Realistic intervention
make sure no one smokes	reduce tobacco consumption to 20% of current
provide treatment to all patients	treat iff a patient is in critical condition
move a robotic arm to coordinate (x, y, z)	move it to (x, y, z) with a small error
mark all applicants as males	mark all applicants as males on paper

2 Canonical types of regimes

Purpose. Classify interventions by the new mechanism they install.

- **Idle/natural:** $X = f_X(\mathbf{pa}_X, u_X)$ (no intervention).
- **Hard/atomic:** $X = x$, $\text{do}(X = x)$, the type considered so far. Every student gets tutoring.
- **Conditional:** $X = g(\mathbf{w})$ sets X to the output of a function of observable variables $\mathbf{W}$. Students get tutoring iff they have a low GPA.
- **Stochastic:** $X \sim P^*(x \mid \mathbf{w})$ makes X follow a given distribution conditional on $\mathbf{W}$. Students with low GPA enter a raffle for 80% of the spots, other interested students for the remaining 20%.

We write σ_X for such an intervention and $P(\cdot; \sigma_X)$ for the induced distribution.

Changes in regime. The unperturbed world, where data are generated, is the natural regime with graph $\mathcal{G}$. An intervention creates a new regime differing in the mechanisms of the intervened variables. Its graph $\mathcal{G}_{\sigma_X}$ is drawn with a regime indicator (a small square σ_X) pointing into the intervened variables, and edges into X replaced according to the new mechanism.

3 Reduction to known machinery

Purpose. See when soft-intervention effects follow from hard-intervention results.

Can identification of soft (conditional, stochastic) interventions be reduced to that of hard interventions? **A reducible example.** Suppose $P(X = 1 \mid Z = 1) = 0.3$: only 30% of motivated students get tutoring. What if this were doubled, σ_X: $P^*(X = 1 \mid Z = 1) = 0.6$? In $\mathcal{G}_{\sigma_X}$, X's parents remain $\{Z\}$ but its mechanism changes. Then

$$\begin{aligned} P(y; \sigma_X) &= \sum_z P(y \mid z; \sigma_X) P(z; \sigma_X) = \sum_z P(y \mid z; \sigma_X) P(z) \\ &= \sum_{z,x} P(y \mid z, x; \sigma_X) P(x \mid z; \sigma_X) P(z) = \sum_{z,x} P(y \mid z, x; \sigma_X) P^*(x \mid z) P(z) \\ &= \sum_{z,x} P(y \mid z, x) P^*(x \mid z) P(z). \end{aligned}$$

Z is not affected by changing X's mechanism; the new assignment is $P^*(x \mid z)$; and given Z and X, Y responds as in the natural regime. Is a do-calculus-like derivation always sufficient?

An intentionally wrong derivation. A patient receives treatment X_1; a second physician observes $Z = z$ and decides a second treatment X_2; finally the patient survives or not (Y). Graph: $X_1 \to Z$, $X_1 \to X_2$, $X_1 \to Y$, $Z \to X_2$, $X_2 \to Y$, $X_1 \leftrightarrow Z$, $Z \leftrightarrow Y$. Consider fixing $X_1 = x_1$ and $X_2 = g(x_1, z)$, i.e. $\mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))$, or $\sigma_{\mathbf{X}} = \{X_1 = x_1, X_2 = g(x_1, z)\}$. One might write

$$P(y \mid \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) = P(y \mid x_1, \mathrm{do}(X_2 = g(x_1, z))),$$

citing Rule 2 with $(Y \perp_d X_1 \mid X_2)$ in $\mathcal{G}_{\overline{X_2}\underline{X_1}}$. This is wrong: under $\mathrm{do}(X_2 = g(x_1, z))$ the edges into X_2 remain active, so $\mathcal{G}_{\overline{X_2}\underline{X_1}}$ is not the right graph. The check must use $\mathcal{G}_{\sigma_{X_2}\underline{X_1}}$, where separation fails because, given the intervention on X_2, the path $X_1 \leftrightarrow Z \to X_2 \to Y$ is open.

Where do-calculus intuition breaks. Under do-interventions the intervened variables lose all dependence on their parents; soft interventions may keep some or all dependencies on parents, change their distribution, or even add new dependencies.

4 σ-calculus

Purpose. State inference rules for soft interventions that consult both pre- and post-interventional graphs.

Theorem 4.1 (σ-calculus, simplified; Correa and Bareinboim 2020)**.** *For disjoint* $\mathbf{X}, \mathbf{Y}, \mathbf{Z} \subseteq \mathbf{V}$ *and disjoint* $\mathbf{T}, \mathbf{W} \subseteq \mathbf{V} \setminus (\mathbf{Z} \cup \mathbf{Y})$*:*

Rule 1 (insertion/deletion of observations) $P(\mathbf{y} \mid \mathbf{w}, \mathbf{t}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}})$ *if* $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ *in* $\mathcal{G}_{\sigma_{\mathbf{X}}}$.

Rule 2 (change of regimes under observation) $P(\mathbf{y} \mid \mathbf{w}, \mathbf{x}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}, \mathbf{x})$ *if* $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{W})$ *in* $\mathcal{G}_{\sigma_{\mathbf{X}}\underline{\mathbf{X}}}$ *and in* $\mathcal{G}_{\underline{\mathbf{X}}}$.

Rule 3 (change of regimes without observation) $P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w})$ *if* $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{W})$ *in* $\mathcal{G}_{\sigma_{\mathbf{X}}\overline{\mathbf{X(W)}}}$ *and in* $\mathcal{G}_{\overline{\mathbf{X(W)}}}$.

Full version. With an outer regime $\sigma_{\mathbf{X}}$ and a regime change on $\mathbf{Z}$ from $\sigma_{\mathbf{Z}}$ to $\sigma'_{\mathbf{Z}}$:

- Rule 1: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{t}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}}$;

- Rule 2: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\underline{\mathbf{Z}}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\underline{\mathbf{Z}}}$;
- Rule 3: $P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\overline{\mathbf{Z(W)}}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\overline{\mathbf{Z(W)}}}$.

4.1 Comparison with do-calculus

Remark. Under $\mathrm{do}(x)$, X is deterministic, so $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x), w, x)$. Under a non-atomic intervention this is not necessarily so: for $X \to Y$, $P(y; \sigma_X) \neq P(y \mid x; \sigma_X)$ in general.

	do-calculus	σ-calculus
Rule 1	$P(y \mid \mathrm{do}(x), w, t) = P(y \mid \mathrm{do}(x), w)$ if $(Y \perp_d T \mid W, X)_{\mathcal{G}_{\overline{X}}}$	$P(y \mid w, t; \sigma_X) = P(y \mid w; \sigma_X)$ if $(Y \perp_d T \mid W)_{\mathcal{G}_{\sigma_X}}$
Rule 2	$P(y \mid \mathrm{do}(x), w) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\underline{X}}}$	$P(y \mid x, w; \sigma_X) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X \underline{X}}$ and $\mathcal{G}_{\underline{X}}$
Rule 3	$P(y \mid \mathrm{do}(x), w) = P(y \mid w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\overline{X(W)}}}$	$P(y \mid w; \sigma_X) = P(y \mid w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X \overline{X(W)}}$ and $\mathcal{G}_{\overline{X(W)}}$

In Rule 1 the relevant graph depends on the specification of the intervention; in Rules 2 and 3 the independence must hold in both the pre- and post-interventional graphs.

Revisiting Rule 2. For the sequential-treatment example,

$$\begin{aligned} P(y \mid \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) &= P(y \mid x_1, \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) \\ &= P(y \mid x_1; \sigma_{X_1, X_2}) \stackrel{?}{=} P(y \mid x_1; \sigma_{X_2}). \end{aligned}$$

To invoke Rule 2 we must check the absence of a back-door path from X_1 to Y in two graphs: the one where X_2 is set by g ignoring Z, and the one where X_2 still listens to X_1 and Z. It holds only in the first, so the rule is not applicable.

4.2 Another intervention

Resources are limited, so tutoring is focused on those who need it most: students with low GPA must get tutoring, and only they can, i.e. $P^*(X = 1 \mid W = 0) = 1$, $P^*(X = 1 \mid W = 1) = 0$, a conditional intervention ($W = 0$ denoting low GPA). In $\mathcal{G}_{\sigma_X}$, X now depends on W (not on Z):

$$\begin{aligned} P(y; \sigma_X) &= \sum_{w,z,x} P(y \mid x, w, z; \sigma_X) P(x \mid w, z; \sigma_X) P(w, z; \sigma_X) \\ &= \sum_{w,z,x} P(y \mid x, w, z; \sigma_X) P(x \mid w; \sigma_X) P(w, z; \sigma_X) && \text{Rule 1} \\ &= \sum_{w,z,x} P(y \mid x, w, z) P(x \mid w; \sigma_X) P(w, z; \sigma_X) && \text{Rule 2} \\ &= \sum_{w,z,x} P(y \mid x, w, z) P^*(x \mid w) P(w, z) && \text{Rule 3.} \end{aligned}$$

5 C-factors and soft interventions

Purpose. Identify soft-intervention effects systematically with c-factors.

For $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$,

$$P(z, x, y) = \sum_{u_Z} P(z \mid u_Z) P(u_Z) \sum_{u_X} P(x \mid z, u_X) P(u_X) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y) = Q[Z]\, Q[X]\, Q[Y].$$

Under $\mathcal{M}' = \mathcal{M}_{\sigma_X}$ with diagram $\mathcal{G}' = \mathcal{G}_{\sigma_X}$, only X's mechanism changes:

$$P(z,x,y;\sigma_X) = \sum_{u_Z} P(z \mid u_Z)P(u_Z) \sum_{u'_X} P'(x \mid z, u'_X)P(u'_X) \sum_{u_Y} P(y \mid x, z, u_Y)P(u_Y)$$

$$= Q[Z]\, Q[X;\sigma_X]\, Q[Y].$$

In general, $P(\mathbf{y};\sigma_{\mathbf{X}}) = \sum_{\mathbf{d}\setminus\mathbf{y}} \prod_{\mathbf{D}_i \in \mathcal{C}(\mathbf{D})} Q[\mathbf{D}_i]$, the product over the c-components of $\mathcal{G}_{\sigma_{\mathbf{X}}}$ restricted to the relevant ancestors; $Q[\mathbf{X};\sigma_{\mathbf{X}}]$ is given by the definition of $\sigma_{\mathbf{X}}$, and every other factor must be identified from $P(\mathbf{V})$, e.g. with IDENTIFY (Lecture 7).

Example 5.1 (Systematic identification 1)**.** $\mathcal{G}$: $W \to Z$, $X \to Z \to Y$, $X \leftrightarrow Y$; intervention $\sigma_X = P^*(X \mid W)$, so $\mathcal{G}_{\sigma_X}$ has $W \to X$ and loses $X \leftrightarrow Y$. C-components: in $\mathcal{G}$, $\mathbf{C}_1 = \{X, Y\}$, $\mathbf{C}_2 = \{W\}$, $\mathbf{C}_3 = \{Z\}$; in $\mathcal{G}_{\sigma_X}$, $\mathbf{D}_1 = \{Y\}$, $\mathbf{D}_2 = \{W\}$, $\mathbf{D}_3 = \{Z\}$, $\mathbf{D}_4 = \{X\}$. Subgoal: identify each $Q[\mathbf{D}_i]$; $Q[\mathbf{D}_4] = Q[X;\sigma_X]$ is given by the intervention. IDENTIFY$(\{Y\}, \{X,Y\}, Q[T], \mathcal{G})$: $A = An(\{Y\})_{\mathcal{G}[\{X,Y\}]} = \{Y\} = C$, so $Q[Y] = \sum_x Q[X,Y]$. IDENTIFY$(\{W\}, \{W\}, \cdot)$ and IDENTIFY$(\{Z\}, \{Z\}, \cdot)$ return immediately. Hence

$$P(y;\sigma_X) = \sum_{w,x,z} Q[X;\sigma_X]\, Q[Z]\, Q[W]\, Q[Y] = \sum_{w,x,z} Q[X;\sigma_X]\, Q[Z]\, Q[W] \sum_{x'} Q[X,Y]$$
$$= \sum_{w,x,z} P^*(x \mid w)\, P(z \mid x, w)\, P(w) \sum_{x'} P(x')\, P(y \mid x', z, w).$$

Example 5.2 (Systematic identification 2: failure)**.** $\mathcal{G}$: $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$; intervention $\sigma_X = P^*(X \mid Z)$. C-components: in $\mathcal{G}$, $\mathbf{C}_1 = \{W, X, Y\}$, $\mathbf{C}_2 = \{Z\}$; in $\mathcal{G}_{\sigma_X}$, $\mathbf{D}_1 = \{Y, W\}$, $\mathbf{D}_2 = \{X\}$, $\mathbf{D}_3 = \{Z\}$. IDENTIFY$(\{W,Y\}, \{W,X,Y\}, Q[T], \mathcal{G})$: $A = An(\{W,Y\})_{\mathcal{G}[\{W,X,Y\}]} = \{W,X,Y\} \neq C$ and $A = T$, so FAIL. (Recall $Q[\mathbf{C}_j] = \prod_{V_i \in \mathbf{C}_j} Q[\mathbf{V}^{\leq i}]/Q[\mathbf{V}^{\leq i-1}]$ and $Q[\mathbf{V}^{\leq i}] = \sum_{\mathbf{v}^{>i}} Q[\mathbf{V}]$.)

6 Summary

- In decision-making settings, where interventions must be implemented, non-atomic (conditional and stochastic) interventions are needed.
- In the spirit of do-calculus, analogous rules consulting both pre- and post-interventional graphs identify the effects of soft interventions.
- Soft-intervention effects can also be identified by systematic procedures based on c-factors.

7 Source coverage

The source deck has 51 physical pages (29 numbered slides with build-ups).

Topic	Physical pages
Motivation: tutoring example, realisable interventions	1–9
Canonical regimes, regime changes	10–15
Reduction, reducible example, wrong derivation, limits of intuition	16–25
σ-calculus and comparison with do-calculus	26–36
C-factors and systematic identification, summary, full rules	37–51